EXTRAI-3 (GEMMA-SÓ): a meta-analysis by a single 12-billion-parameter local model, under a harness that only warns

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Abstract

The companion studies established that small local language models read clinical trials nearly perfectly but fail at orchestration, and that moving everything deterministic into code isolates reading as the whole game [Barra, 2026a,b]. This third study asks the inverse question, pre-registered before any run: *what is the MINIMAL harness under which the best measured reader — `gemma4:12b`, quantized, on an integrated GPU — orchestrates the meta-analytic arithmetic of its own extraction sheets?* The design constraint that keeps the question meaningful: harness nets may **detect and warn, never substitute a value**. Climbing a pre-registered ladder of detection-only nets (sign echo; schema-constrained calls with declared per-argument sources; an argument type system; coherence checks; a closing-restatement check; an escalation flag — ten nets at freeze), the model assembled per-study calls at up to 6/7 exactness, corrected itself when warned (a swapped-bounds negative SD; a pure copy error in its own closing), and **executed the DerSimonian–Laird pooling over its own sextets, faithfully and digit-consistently, three times in a row** — the call no model of any size had executed in the companion benchmark. Three complete pipelines then ran extraction-to-result; across three independent fresh extractions the sealed unperturbation lens landed on the published value every time (-0.24 $[-0.33, -0.16]$ vs. -0.24 $[-0.32, -0.16]$), while per-study orchestration oscillated (5/7; 6/7 plus one flagged; 5/7 plus one starved) under an identical frozen harness: **reading is reliable; orchestration is weather.** The record doubles as an instrument study: five harness errata, each transcript-proven — including nets that steered correct values into corruption before an amendment fixed them — one erratum caught by the author reading a figure, and a final run in which every link behaved honestly while the composition still distorted (a faithful pool of six studies dominated by one study's tiny stored dispersions), motivating product-level flags. An orchestrator exchange under the same harness showed that code specialization moves the blind spots rather than the score — a same-base code fine-tune tied at 5/7 with complementary failures (their union: 6/7), while an older-generation code specialist emitted flawlessly well-formed JSON and assembled not one valid interval: *form is not semantics* — and a truth-free two-orchestrator committee auto-accepted only exact results, flagging everything else. Harness complexity is reported as a cost metric (1→2→4→8→10 nets). All protocols, eight dated amendments, prompts, transcripts of every turn — including all invalidated runs — graders and products are public and archived.

1 Introduction

The first EXTRAI study measured four small local models as near-perfect readers and failed orchestrators; the second moved everything deterministic into pre-registered code and showed that, with reading as the only model stage, exactly one iGPU-scale extractor reconstructs a

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published meta-analysis end to end [Barra, 2026a,b]. The natural response for production is the second study's architecture. The natural response for *science* is the inverse experiment, and its author ordered it directly: take the best reader and try to make it the orchestrator — correcting stage by stage — of a meta-analysis pipeline that is 100% one model.

That experiment is only meaningful under a self-denying rule, frozen in the protocol before any run: **the harness may detect and warn, never substitute**. A net that fixes an argument turns the study back into the companion's code-assembled architecture and kills the claim "the model orchestrated". Everything reported here — including the failures, the starvations, and the harness's own five errata — happened under that rule.

The study contributes: (1) a pre-registered **ladder of detection-only nets**, each rung admitted only against a named, measured failure, with **harness complexity reported as a cost metric**; (2) the benchmark's first measured **model-executed pooling**, reproduced three times; (3) three complete single-model pipelines, extraction to forest plot, whose **unperturbation lenses land on the published value in all three independent fresh extractions**; (4) the measured **asymmetry** between reading (stable) and orchestration (run-to-run weather) under an identical frozen harness; (5) an instrument record in the benchmark's tradition — five harness errata, transcript-proven, two of them nets that damaged correct behavior before amendment — and the resulting doctrine: **detection can refuse, it cannot teach; and local honesty at every link does not compose into global sanity.**

2 Inherited constitution, and the rule

Corpus, anchor, sealed perturbations, the frozen Portuguese extraction instrument, two-layer source-verified key, mechanical grading and the adjudication rite are the companion studies', unchanged [Barra, 2026a]; the anchor's published diamond (-0.24 $[-0.32, -0.16]$, low-carbohydrate diets and HbA1c [Khraise, 2026]) remains the external reference through the graders' sealed unperturbation lens. The model under test is the pinned `gemma4:12b` build, reasoning channel off, on a consumer mini-PC's integrated GPU; the calculator is the companion's function set (per-study mean difference and CI; CI$\to$SD, SE$\to$SD and Cochrane r=0.5 conversions [Higgins et al., 2023]; DerSimonian–Laird pooling [DerSimonian and Laird, 1986]), executed by the harness *on the model's command.* One call per turn; every warning, re-emission, insistence and closing is archived verbatim. Inputs for the orchestration rungs are the model's own extraction sheets; each complete pipeline re-extracts from zero.

3 The ladder

Each rung ran only after the previous was graded; every rung's full transcripts — including the two runs invalidated by harness defects — are archived.

G0 (baseline, measured in the companion). Free-text calculator calls under flow nets: workflow succeeds, content fails — signs dropped in argument assembly, a wrong-side diamond (+0.37). One net.

G1 — the sign-echo net (two nets). One addition: an argument whose magnitude exists in the sheet with the opposite sign, while no sheet number equals it as emitted, draws a warning naming the field. Result: **5/7 studies exact** against the graders' truth over the same sheets, in 5.2 minutes. The Dorans transcript is the rung's anatomy: the model swapped CI bounds, producing a *negative* SD, assembled the effect call anyway, was warned, stubbornly repeated once, then corrected — and closed exact. **The model corrects itself when told where to look.**

G2 (invalidated) and G2b — structured calls (four nets). Calls became JSON under schema-constrained decoding (function name from an enum; numeric arguments; a declared source field per argument), adding a source-check net and an arity net. The first G2 run is preserved as **evidence of instrument-induced damage**: the source net resolved declared fields arm-blind, fired on *correct* control-arm values, and steered the model to corrupt them — the harness did to this model what the companion study's leading question had done to another; separately, a five-argument call perseverated against a truncated foreign-language error because no arity net existed. Both errata were amended (arm-agnostic source check; arity warnings carrying the full Portuguese signature), and G2b re-ran: **5/7, zero format failures, zero spurious warnings** — transcription solved; what remains are *derived* values.

G3 (invalidated) and G3b — the pooling (the deciding rung). The single conversation the companion benchmark's models always failed: emit and execute the pooling call over one's own per-study sextets. The first run is again instrument evidence: the schema helper had bound the *previous rung's* grammar, so constrained decoding still admitted per-study functions, and an off-menu call was executed as an empty pool returning a meaningless division-by-zero; the model reasonably perseverated against an incoherent harness. Amended (per-rung grammar binding; guidance instead of meaningless execution), G3b **passed in one turn**: all seven of its own sextets faithfully transcribed at first emission, the aggregate executed and closed **consistent to the digit** (-0.50 [-0.74, -0.26]). Under the correct grammar, off-menu emissions are unrepresentable — and **a 12B model executed a meta-analytic pooling for the first time in this benchmark's record.**

G2c — typed calls (eight nets). The residual failure class had one shape: **role errors with honest provenance** — printed levels passed as change means; the study's own result passed as an arm mean; correctly derived SDs abandoned for raw ones. Every provenance check passes on such calls, because the declared origins are truthful; only the *role* is wrong. G2c added an argument *type system* (each function slot declares the class of sheet field that may feed it), coherence checks (an interval must contain its own MD; an SD cannot be negative; an n must be an integer above 1), a closing-restatement check (the model's final answer must match its own executed results — transcription is precisely where it fails), and an escalation flag. Result: **6/7 exact — the program's best — plus the seventh arriving FLAGGED, not silent**, on the flag's maiden run: the model swapped-and-unsigned one CI's bounds (caught as a negative SD at the effect call; corrected to the right magnitude), then committed a pure copy error in its closing (`ic95 [0.08, 0.08]`; caught by the restatement net; corrected), and its final differs from the graders only by a defensible dispersion route — exactly what human review is for. The registered prediction paid where it mattered: **the type net converted the levels-as-means habit into a warned self-correction, and the Goday trial closed on the correct route — digit-exact — for the first time in the whole program.**

4 Three pipelines, extraction to result

v1 — from the existing sheets. All stages one model: per-study orchestration (7/7 closed, zero warnings needed), pooling **consistent again** (-0.46 [-0.67, -0.26], digit-exact vs. its own sextets), participant totals pre-computed by code (the companion's lesson: no mental sums), and a Portuguese narrative synthesis with **zero orphan numbers** that correctly singled out the only trial whose interval crosses zero. A deterministic script drew the forest.

v2 — from fresh extraction, with the roadmap's first improvements. Stage E re-ran from zero under schema-constrained decoding (the frozen prompt verbatim; malformed sheets

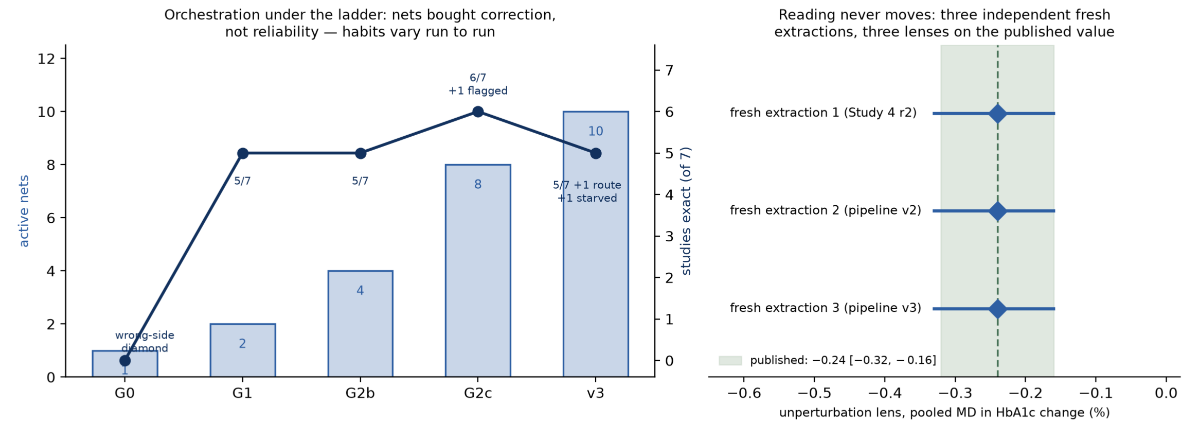

Figure 1: The study in one view. **Left**: harness complexity (active detection-only nets, bars) against per-study orchestration outcome (line) across the ladder and the final frozen-harness pipeline — nets bought self-correction, not reliability; the same frozen 10-net harness yielded 6/7 in one run and a starved study in the next, because the model's habits vary between runs. **Right**: what does not vary — three fully independent fresh extractions (Study 4's round 2; pipeline v2; pipeline v3), three sealed unperturbation lenses, three landings on the published value (-0.24 $[-0.33, -0.16]$ each time, vs. published -0.24 $[-0.32, -0.16]$, shaded). Reading is reliable; orchestration is weather.

unrepresentable) with a *runtime anti-invention net*: every sheet number absent from the source text draws one question, article in context, the model's answer applied as its own correction. Measured: **zero triggers** — none of this extractor's numbers is absent from its source, which is itself the measurement (the net exists for the invention class other models exhibited [Barra, 2026b]). Downstream, pooling consistent a third time, Δ to the graders' truth 0.01 — and **the lens landed on the published value:** -0.24 $[-0.33, -0.16]$. The run also delivered erratum E5-5, **caught by the author reading the figure**: the Goday closing carried an interval that does not contain its own MD ($[-9.22, -8.58]$ around an MD of -1.8 — the model had passed its previous *result* as an arm mean, provenance honestly declared), and the product layer had endorsed the reported final: the forest plotted it, the synthesis cited it, and the orphan scan was structurally blind to provided garbage. The fix is a product doctrine: **flag incoherent reported values, never endorse them** — implemented, with the coherence check promoted into the ladder (G2c) and the whole episode a live demonstration that mechanical checks see only what they were designed to see; the design invites precisely the human reading that caught this.

v3 — the frozen harness, and the honest ceiling. Two micro-nets from G2c's residue (bounds-order warned at the conversion call; a negative conversion result returns with an attached warning), the harness **declared frozen at ten nets**, and everything re-ran from a third fresh extraction. The registered 7/7 prediction was **refuted, in the instructive direction**. The lens hit the published value a *third* time. But the over-conversion habit — absent in v1's calc, present in G2b's, absent in v2's — returned: on Goday the model again derived the change-SD correctly and then insisted on re-converting its own result; the nets correctly refused every wrong move, seven times, and the study starved at the turn cap, closing null and leaving the pool. **Detection can refuse; it cannot teach.** (Starvation is still the right failure mode: an absent, visible result rather than a hidden wrong one.) And the round's second lesson: the surviving six were pooled *faithfully* — and the aggregate was distorted anyway (-0.19 $[-0.23, -0.16]$, I^2 0%), because without the largest effect, one study's tiny stored dispersions took dominant weight. **Local honesty at every link does not compose into global sanity**; the registered successors are product-level flags (a study with sheets but no pooled row; any single study above a declared share of the pooled weight), not more nets

Table 1: The orchestrator exchange: same sheets, same frozen 10-net harness, three orchestrators. "Exact" = final MD and both CI bounds within ±0.01 of the graders' mechanical truth over the same sheets.

Orchestrator	Exact /7	Pooling	Where it fails
`gemma4:12b` (base; the reader)	5	consistent (×3)	a private dispersion route; Goday starved
`codegemma` (older-gen code)	**0**	—	every CI: arities, ns in bound slots
`gemma-4-12B-coder` (same base)	5	consistent	a dispersion mess (Wang); Goday starved

on the model.

5 The orchestrator exchange, and the first committee

Three orchestrators, one frozen harness. By the author's directive, the reader was kept (the same fresh sheets) and the orchestrator swapped: the base model against an older-generation code specialist (`codegemma`) and against a community code fine-tune of the *same* base and size (`gemma-4-12B-coder`; unofficial provenance, digest-pinned). Identical ten nets, identical instruments; every final re-graded mechanically against the sheets' truths.

Three results the exchange forces (Table 1). First, **generation and family instruction-tuning dominate "code" specialization**: under schema-constrained decoding the older code model emits flawlessly well-formed JSON and supplies no statistical role semantics at all — it copied every mean correctly and assembled not one valid interval. *Form is not semantics.* Second, **the same-base code delta does not change the score; it moves the blind spots**: the coder takes the graders' dispersion route on the study where the base keeps a private one, and fumbles the study the base handles cleanly — the union of the two covers 6/7, everything but Goday, echoing at the orchestration layer the complementary-blind-spots finding the companion benchmark measured between auditing models [Barra, 2026a]. Third, **the Goday derivation wall is architectural, not a specialization gap**: both same-generation 12B models starve on it identically, under identical warnings. The pooling call, meanwhile, was executed digit-consistently by *both* family models — four consistent poolings across two orchestrators.

The committee, measured at zero model cost. The union begged the committee test, and its rule needs no truth: per study, if both orchestrators' finals agree, auto-accept; if one is null, the other stands flagged *single-source*; if both are present but different, flag *discordant* for a human. Composed mechanically over the two archived transcripts, the committee auto-accepted four studies — **all four exact, the registered prediction** — and flagged the other three (two discordant, one absent), passing **zero wrong values without a flag**. A truth-free agreement rule over two cheap, blind-spot-complementary orchestrators bought perfect precision where it acted alone and honest escalation everywhere else. The same amendment's product flags, run retroactively, convert the frozen-harness pipeline's silent composition distortion into two named warnings (a missing study; one study at 79.7% of the pooled weight against a 40% threshold).

6 Discussion

The asymmetry is the finding. Under one frozen harness, one model, one corpus: three independent fresh extractions put the sealed lens on the published value three times, while per-study orchestration returned 5/7, then 6/7 with a flag, then 5/7 with a starvation — with individual habits (over-conversion; bounds-swapping) appearing and vanishing between runs.

Reading, the stage the companion study isolated as the whole game, is reproducibly anchor-faithful; orchestration at 12B is weather. For production the implication is unchanged and now double-measured: keep everything deterministic in code [Barra, 2026b]. For capability research, GEMMA-SÓ is the measured map of the frontier — including the benchmark's first model-executed pooling, reproduced three times, which moves one stage from "cannot" to "can, consistently".

What would move the ceiling. Four remedies, ordered by how much of the "the model did it" claim each preserves, all pre-registrable: (1) **committee orchestration with an agreement rule** — and, measured, the committee specifically rather than a replicate: a pre-registered replicate arm found the starving habit *sheet-conditioned*, not run-stochastic (four runs on the same sheet all starved, while a different extraction round's sheet had run clean — our earlier weather framing had confounded sheet with run), so a second run of the same model neither rescues the starved study nor catches a consistent private route it happily agrees with; a second *orchestrator* with complementary blind spots does change the game (four auto-accepts, four exact, zero unflagged errors — Section 5); (2) a **temperature-zero arm** — also already measured: greedy twins were turn-identical in only 5/7 studies and bought no exactness, so the residual variance is not sampling temperature; (3) **second-level pedagogical warnings** — after repeated refusals of one class, a frozen warning text may state the rule itself ("an SD derived by r=0.5 is already an SD"), teaching by instruction without touching values; (4) a **deterministic floor with labels** — a study that exhausts its budget is computed by code and enters the product marked as such, converting starvation into graceful, honest degradation ("six by the model, one by code, labeled"). The fourth is the production hybrid: the companion architecture serving as this study's safety net.

Instruments are the other subject. Five harness errata in one study, all transcript-proven, two of them nets that actively damaged correct behavior until amended — the arm-blind source check steering true control-arm values toward corruption is the same phenomenon as the companion study's leading trigger question, now committed by code. The discipline that made this survivable is the benchmark's standing doctrine: dated amendments before re-runs, defective runs preserved as evidence, the adjudicator's own errors published — and, new here, **harness complexity reported as a cost** (1, 2, 4, 8, 10 nets for 0, 5, 5, 6, 5 exact studies): a curve that disciplines the temptation to fix every failure with one more net, since its flat segments are visible.

The human is a stage, not a spectator. The orphan scan passed a garbage interval because the garbage was in the data it was given; the author, reading the figure, caught it. Mechanical verification sees what it was designed to see; the escalation flag exists to *summon* the human look rather than depend on its luck. In a study whose thesis is a self-limiting harness, the measured necessity of occasional human adjudication is not a concession — it is the position of the evidence-synthesis organizations [Flemyng et al., 2025], arrived at from the instrument side.

7 Limitations

One model, one corpus, one continuous outcome, one quantization; every claim is scoped to them. Orchestration reliability estimates rest on single runs per rung (the replicate remedy is registered, not yet run). The two residual per-study divergences are dispersion-route choices — internally correct arithmetic on defensible readings — whose acceptability belongs to the answer key's documented-routes mechanism, not to the harness. The warning texts, though frozen and detection-only, carry design choices of the same human–AI pair that graded the runs, mitigated by verbatim transcripts and published errata rather than independence [Zheng et al., 2023]. The pooled products of the single-model pipelines are demonstrations of orchestration, not substitutes for the deterministic architecture's estimates. Generalization to a new anchor remains the benchmark's designed, pre-registered next step.

8 Conclusion

Given a harness that only warns, a 12-billion-parameter local model read seven trials faithfully three separate times, assembled its own calculator calls at up to six of seven studies exact, corrected its own sign flips and copy errors when told where to look, and executed the statistical synthesis of a meta-analysis — the step no model in this benchmark had ever performed — three times, consistently. The same record shows what it cannot yet own: habits that come and go between runs, a vice that warnings can block but not cure, and compositions that can distort while every part behaves. The ladder's cost curve, the frozen nets, the starved study and the three unmoved lenses are, together, the measured answer to this study's question: the minimal harness exists, it is small, it is now frozen — and what it buys is a reader that can also, on its good days, be the statistician; on the other days, the code and the human it summons are still the difference between an absent result and a wrong one.

Data and code availability

All protocols (pre-registered, with eight dated amendments), frozen prompts and warning texts, the harness with every net, full turn-by-turn transcripts of every rung and pipeline — including both invalidated runs — graders, adjudications, products (forests, syntheses) and the exact pinned model build are public and archived at Zenodo [Barra, 2026c]:
`https://github.com/gbbarra/llm-evidence-extraction-benchmark-ptbr`. Closed-stratum primary articles are not redistributed; scripts rebuild the corpus from legally obtained copies.

Declarations

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Clinical trial registration. Not applicable: no clinical trial was conducted. The randomized trials analyzed are published reports registered by their original investigators.

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